

Sinan Bilir

ARTIFICIAL INTELLIGENCE & DATA SCIENTIST | MLOps, NLP, Generative AI & Cloud Automation

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PROFESSIONAL SUMMARY

Artificial Intelligence and Data Scientist with an MSc in Data Science and Artificial Intelligence (Distinction) and nine years of analytical and commercial experience. Specialises in applied machine learning, deep learning, natural language processing (NLP), generative AI, and business intelligence. Builds end-to-end solutions spanning data preparation, feature engineering, model development, SHAP-based explainability, and MLOps deployment. Current work includes NeoJan, a neonatal risk prediction platform integrating multi-omics and synthetic clinical data with NHS digital health stakeholders, achieving a PR AUC of 1.0. Commercial experience includes developing machine learning models that contributed to a 20% increase in job applications and data-driven strategies that generated a 15% increase in sales revenue. Deploys reliable, scalable machine learning systems across Azure, AWS, and Google Cloud Platform using Docker, MLflow, and CI/CD pipelines in research & commercial environments.

SKILLS

Programming, Core Tools & Languages: Python, SQL, Bash, R, Git, GitHub, GitLab, Terraform, Agile, LaTeX, Jupyter Notebook.

Machine Learning & AI Architectures: Scikit-learn, XGBoost, LightGBM, TensorFlow, Keras, PyTorch, Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory networks (LSTMs), Generative Adversarial Networks (GANs), Autoencoders, Transfer Learning, SHAP, LIME, Reinforcement Learning, Federated Learning.

Natural Language Processing (NLP) & Generative AI: SpaCy, NLTK, Hugging Face Transformers, BERT, GPT-based architectures, Large Language Models (LLMs), Text Classification, Semantic Similarity, Topic Modelling, Information Extraction Pipelines, Prompt Engineering.

Cloud, MLOps & Deployment: Microsoft Azure Machine Learning, Amazon Web Services (AWS: S3, EC2, Lambda, SageMaker), Google Cloud Platform (GCP: BigQuery, Vertex AI), Docker, Kubernetes, MLflow, Apache Airflow, TensorFlow Serving, CI/CD (GitHub Actions, Jenkins), Model Versioning, Model Monitoring, Retraining Pipelines, Flask, FastAPI, REST APIs.

Data Engineering & Big Data: Apache Spark, Hadoop, Databricks, MySQL, PostgreSQL, MongoDB, Extract Transform Load (ETL), Feature Engineering, Orchestration, Genomics-oriented Data Processing, Multi-omics Integration, Synthetic Clinical Data.

Analytics, Visualisation & Business Intelligence: Matplotlib, Seaborn, Plotly, Power BI, Tableau, Looker, Streamlit, Dash, A/B Testing, Statistical Analysis, Business Analysis, Business Intelligence, Sales Engineering.

Evaluation & Performance Metrics: ROC AUC, PR AUC, Precision-Recall, F1 Score, Brier Score, Calibration Analysis, Model Explainability, Cross-Validation Strategies.

WORK HISTORY

ML and AI Systems Multi-Omics Integration Lead Researcher

University of Liverpool | 12/2025 – 08/2026 | Remote

- Lead an interdisciplinary research initiative spanning machine learning, healthcare AI, and biomedical data science, with direct relevance to regulated, risk-sensitive decision-support environments.
- Designed and implemented an end-to-end data and machine learning pipeline integrating gene expression datasets (GSE122340, GSE46995), metabolomics repositories (Metabolites, Metabolomics Workbench), and synthetic electronic health records from Synthea.
- Engineered and evaluated Random Forest, XGBoost, and deep neural network models for neonatal risk prediction, increasing recall by 15% and optimizing clinical decision support.
- Engineered high-dimensional clinical datasets using clustering and correlation analysis, uncovering novel cross-domain insights to support biomarker identification.
- Built and deployed the NeoJan Streamlit application with structured CSV/JSON ingestion, interactive 3D PCA visualisation, dynamic threshold tuning, and automated clinical-style reporting.
- Integrated SHAP-based explainability to improve transparency, auditability, and clinician interpretability of model predictions.
- Architected Dockerized environments, MLflow tracking, and CI/CD pipelines to scale research operations and ensure 100% model reproducibility.
- Collaborated within ethics-approved frameworks involving NHS digital health stakeholders and external healthcare research partners, with ongoing preparation for pilot deployment in NHS partner settings.

Data Scientist (Contract)

MGL Licensing | 02/2024 – 05/2024 | Remote

- Conducted customer segmentation using K-Means clustering to identify behavioural patterns and support retention-focused interventions.
- Designed and delivered interactive analytical dashboards to support operational monitoring and strategic decision-making.
- Built near real-time analytics workflows to surface behavioural trends, performance bottlenecks, and commercial opportunities.
- Applied A/B testing methodologies to optimise engagement strategies and improve return on investment.

Data Scientist (Contract)

British Airways | 03/2024 – 05/2024 | Remote

- Developed a Random Forest classifier to predict customer booking behaviour, improving understanding of conversion drivers and likely booking outcomes.
- Performed exploratory data analysis and engineered high-impact features to strengthen model performance.
- Communicated analytical findings and business recommendations clearly for non-technical stakeholders.

Lead Sales Operations Business Analyst

Antal TECH Jobs | 08/2022 – 04/2023 | Greater London, UK

- Built and applied machine learning models to predict job-seeker engagement, contributing to a 20% increase in applications.
- Designed automated dashboards using Power BI and SQL to improve recruitment reporting and decision-making.
- Used Google Analytics and A/B testing to optimise marketing strategy, contributing to a 15% improvement in conversion rates.

Business Intelligence Consultant

SB Sales and Marketing Consultancy Ltd | 04/2019 – 03/2022 | Greater London, UK

- Designed and managed business intelligence solutions delivering actionable insight into sales performance and market trends.
- Applied predictive modelling in Python to support forecasting, scenario analysis, and business strategy optimisation.
- Wrote complex SQL queries to extract, transform, and analyse key metrics for commercial planning and performance review.

EMEA Senior Sales Engineer and Data Analyst

Ant Kalip M&G Group | 02/2017 – 02/2019 | Kocaeli, Turkey

- Analysed sales trends, KPIs, and operational performance using SAP ERP and Excel, contributing to a 15% increase in sales efficiency.
- Utilized customer behaviours data and Google Analytics to monitor market performance, boosting campaign efficiency and refining global product positioning.
- Created Power BI dashboards and Excel reports to improve visibility of commercial and operational metrics and support data-driven decisions

PROJECTS

NeoJan – Multi-Omics Machine Learning for Neonatal Jaundice Prediction | University of Liverpool | MSc Data Science & Artificial Intelligence

- Developed the world's first integrated multi-omics machine-learning framework for neonatal jaundice risk prediction. Unified omics + clinical data using robust preprocessing and harmonisation (SCANPY, MetaboAnalystR, biomaRt). Constructed and compared Random Forest, XGBoost, and Deep Neural Network models, achieving PR AUC = 1.0. Embedded explainable AI modules (SHAP/LIME) for transparency and clinician trust. Created a clinician-oriented Streamlit web application allowing real-time patient assessment, threshold visualisation, and decision-support aligned with NHS guidelines. Deployed via Docker + MLflow, ensuring reproducibility, security, and CI/CD integration. Technologies: Python, Scikit-Learn, TensorFlow, Pandas, Streamlit, Plotly, Docker, MLflow, GitHub. Collaborators: NHS Digital Health | University of Liverpool | PhysioNet.

Pneumonia Detection Using Convolutional Neural Networks | University of Liverpool | MSc Data Science & Artificial Intelligence

- Developed and deployed a CNN-based deep learning model for pneumonia detection using chest X-ray images, leveraging TensorFlow and Keras. Implemented data preprocessing (normalization, augmentation) and addressed class imbalance using class weight adjustment techniques. Applied Transfer Learning (ResNet15) and K-Fold Cross-Validation, improving model generalization and achieving 97.6% accuracy & 98% recall. Integrated model interpretability techniques (Grad-CAM, SHAP) to enhance explainability for healthcare professionals.

Deep Learning Fashion MNIST Classification Project | University of Liverpool | MSc Data Science & Artificial Intelligence

- Developed and evaluated deep neural network models for Fashion-MNIST image classification, comparing Sigmoid and ReLU activation functions and Dropout regularisation to optimise convergence and reduce overfitting. Trained and benchmarked models using accuracy and loss metrics, demonstrating practical application of deep learning, neural networks, model optimisation, and performance evaluation.

Predicting Atmospheric Emissions Using Machine Learning Models | University of Liverpool | MSc Data Science & Artificial Intelligence

- Developed predictive models to estimate nitrogen oxides (NOx) and particulate matter (PM2.5) emissions across London. Utilized geospatial data from the London Atmospheric Emissions Inventory (LAEI 2019) and applied machine learning algorithms, including Random Forest and XGBoost. Implemented feature engineering techniques such as Principal Component Analysis (PCA) for dimensionality reduction. Technologies: Python, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, GeoPandas, Jupyter Notebook

Data Quality Analysis and Cleansing of SIS Faculty Dataset | University of Liverpool | MSc Data Science & Artificial Intelligence

- Performed data quality assessment, cleansing, and preprocessing on a Student Information System (SIS) faculty dataset, resolving missing values, inconsistent formatting, duplicate records, and data quality issues to improve dataset reliability and prepare structured data for machine learning and analytical workflows. Technologies: Python, Pandas, Jupyter Notebook.

Secure Client-Server Communication System | University of Liverpool | MSc Data Science & Artificial Intelligence

- Developed a secure client-server communication system in Python, emphasizing secure data exchange using symmetric encryption. Supported multiple data serialization formats, including JSON, Binary (Pickle), and XML. Designed with modularity and extensibility in mind. Technologies: Python, Fernet (symmetric encryption), JSON, Pickle.

Shortest Path Algorithm Implementation | University of Liverpool | MSc Data Science & Artificial Intelligence

- Implemented and tested Floyd-Warshall's Algorithm in Python to compute all-pairs shortest paths in weighted graphs, analysing time and space complexity, algorithmic efficiency, and graph traversal performance. Technologies: Python, Graph Algorithms, Data Structures.

EDUCATION

Master of Science (MSc) in Data Science & Artificial Intelligence

University of Liverpool | Oct 2023 – Apr 2026 | Distinction

Relevant study: Machine Learning in Practice, Deep Learning, Software Development in Practice, Databases and IT Systems, Mathematics and Statistics for AI & Data Science, Cloud Computing;

Dissertation: Development of a Machine Learning Model & Software Platform for Multi-Omics-Based Prediction of Neonatal Jaundice Risk.

Bachelor of Science (BSc) in Industrial Engineering

Dokuz Eylul University | Sep 2011 – Aug 2016 | Upper Second-Class

Relevant study: Operations Research, Production Systems, Statistics, Manufacturing Processes, Quality Planning and Control, Simulation, Optimisation, Logistics, Supply Chain Management, Project Management, Information Systems; **Thesis:** Transition to Just-in-Time Production and Kanban Board Implementation in an Automotive Supplier Firm. Leadership: Regional Manager, ESTIEM, leading international projects in business process optimisation and supply chain management.

RESEARCH & PUBLICATIONS

- Bilir, S. (2016). Just-in-Time Production and Kanban Board Implementation. BSc Dissertation, Dokuz Eylul University.
- Bilir, S., Garron, N., Kelm, K. (2025). Multi-Omics-Based Prediction of Neonatal Jaundice Risk. University of Liverpool; target journal: Frontiers in AI (2026).

CERTIFICATIONS

Artificial Intelligence Programme – University of Oxford (In Progress); Machine Learning with Scikit-Learn – LinkedIn (Sep 2024); Business Analysis Foundations – IIBA (Aug 2024); Business Analysis: Essential Tools and Techniques – Project Management Institute; Machine Learning with Python: Foundations – LinkedIn (Jan 2024).